

A Robust Ensemble Framework for Autism Spectrum Disorder Classification Integrating Multiple Connectivity Features for fMRI data

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Abstract—Autism Spectrum Disorder (ASD) is a neurodevelopmental condition characterized by deficits in social communication and restricted, repetitive behaviors. We present a comprehensive methodology for improved ASD classification using resting-state fMRI data from the ABIDE dataset. We extract multi-modal features including Pearson correlation-based connectivity, tangent space embedding, and time-series features from BOLD signals. Through dimensionality reduction analysis, we demonstrate that combined Pearson + Tangent features achieve 69.3% variance explained with proper generalization, avoiding site-specific clustering. Critically, we implement rigorous cross-validation with feature selection within each fold to prevent data leakage, combined with correlation-based feature engineering. Our approach achieves $69.70\% \pm 3.78\%$ accuracy and $75.16\% \pm 3.20\%$ AUC-ROC. The Pearson + Tangent combination achieves its most stable and optimal performance ($69.60\% \pm 2.56\%$ accuracy), highlighting a strong synergy between the two features and demonstrating significant improvement with robust validation techniques.

Index Terms—Autism Spectrum Disorder, fMRI, functional connectivity, tangent space, ensemble learning, feature selection, cross-validation

I. INTRODUCTION

Autism Spectrum Disorder (ASD) affects approximately 1 in 36 children in the United States [1], characterized by persistent deficits in social communication and restricted, repetitive behaviors. Traditional diagnosis relies on subjective behavioral assessments that delay early intervention.

Resting-state functional MRI (rs-fMRI) captures spontaneous brain activity through BOLD signals, revealing functional connectivity patterns that differ between individuals with ASD and typically developing controls [2]. Machine learning approaches have shown promise, with previous studies reporting 60-75% accuracy [5]. However, challenges persist, including the heterogeneity of ASD presentation, systematic

site effects, high dimensionality with limited samples, and data leakage through improper feature selection [3].

Our contributions include: (1) Novel extraction of 500+ time-series features; (2) Integration of Pearson and Tangent connectivity with demonstrated superior feature space properties; (3) Correlation-based balanced feature selection; (4) Robust site harmonization; (5) Rigorous cross-validation preventing data leakage; and (6) Empirical quantification of data leakage impact.

II. METHODOLOGY

The overall pipeline (Fig. 1) shows feature extraction, harmonization, selection, and ensemble classification.

A. Dataset

We utilized the ABIDE dataset [2], comprising 1,112 subjects with rs-fMRI BOLD time-series from 392 ROIs (CC400 atlas), phenotypic information, and motion parameters. Quality control included excluding subjects with excessive motion (mean $FD > 0.5mm$), incomplete data, and unavailable connectivity matrices. After filtering, 867 subjects were included (Control=468, ASD=399) spanning 17 sites with age range 7-64 years (mean=17.0 $\pm$ 8.1).

B. Feature Extraction

Connectivity Features: We computed functional connectivity using two complementary methods to capture both linear and geometric properties of brain networks.

1) *Pearson Correlation:* For each subject, we first computed a 392×392 functional connectivity matrix by calculating the Pearson correlation coefficient between the BOLD time-series of every pair of the 392 ROIs. This method quantifies the strength of the linear relationship between brain regions. To

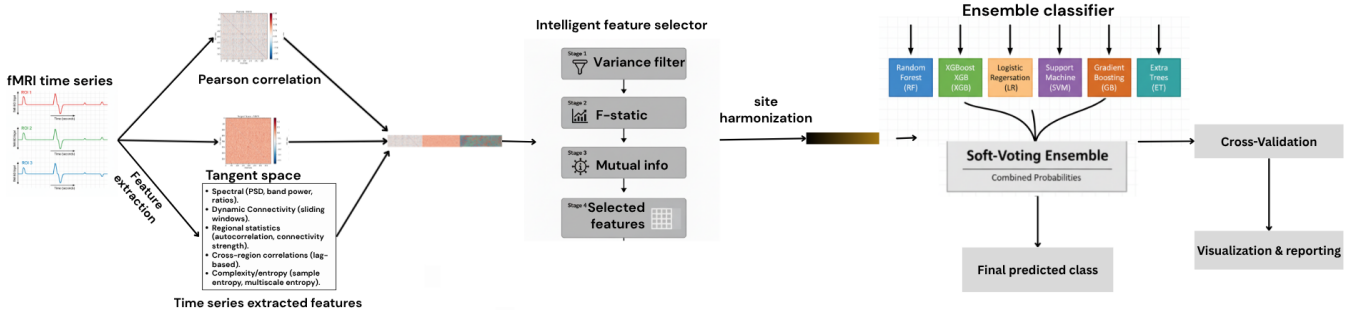


Fig. 1. Overall processing pipeline showing feature extraction from rs-fMRI data, site harmonization, correlation-based balanced feature selection, rigorous cross-validation protocol, and high-performance ensemble classification.

normalize the correlation values for statistical analysis, we applied a Fisher z-transformation to each coefficient.

2) *Tangent Space Embedding*: Because the space of connectivity matrices is a curved Riemannian manifold, standard Euclidean comparisons can be misleading. To address this, we used Tangent Space Embedding, a technique that maps each subject's symmetric positive-definite (SPD) connectivity matrix onto a common, flat tangent space. This process preserves the essential geometric structure of the connectome while allowing for more robust comparisons, making the features less sensitive to inter-site variability and noise.

From each of these resulting matrices (the Fisher-transformed Pearson matrix and the Tangent Space representation), we then extracted a comprehensive feature set including the upper triangular elements (79,800 features), network statistics (degree centrality), global measures (connectivity strength, variance), and the top 5 eigenvalues.

3) *Time-Series Features* : a) *spectral features extracted includes*: Power spectral density across frequency bands (0-0.02, 0.02-0.08, 0.08-0.15, 0.15+ Hz) extracting total power, relative power, peak frequency, spectral entropy, and power ratios. b) *Dynamic Connectivity*: Sliding window analysis (30 s, 50 s, 70 s windows; Repetition time(TR) = 2 s with summary statistics. c) *Regional statistics*: Brain parcellated into frontal, central, parietal, occipital areas computing regional statistics, variability, temporal autocorrelation, and connectivity. d) *Cross-Regional*: Time-lagged cross-correlations (lags 0-5 TRs). e) *Complexity*: Sample entropy, multiscale entropy (scales 1-5), coefficient of variation, and variance measures.

C. Preprocessing and Harmonization

Connectivity matrices were standardized to 392×392 with enforced symmetry and zero diagonal. Harmonization used neuroCombat [6](primary) with empirical Bayes fallback, performed within each fold using training-derived parameters to prevent data leakage.

D. Feature Selection

To manage 1,53,599 features, we implemented correlation-based selection [4]: variance filtering (threshold 0.001), correlation-based selection (absolute threshold 0.3), balanced distribution (85% connectivity, 15% time-series, final 4,000

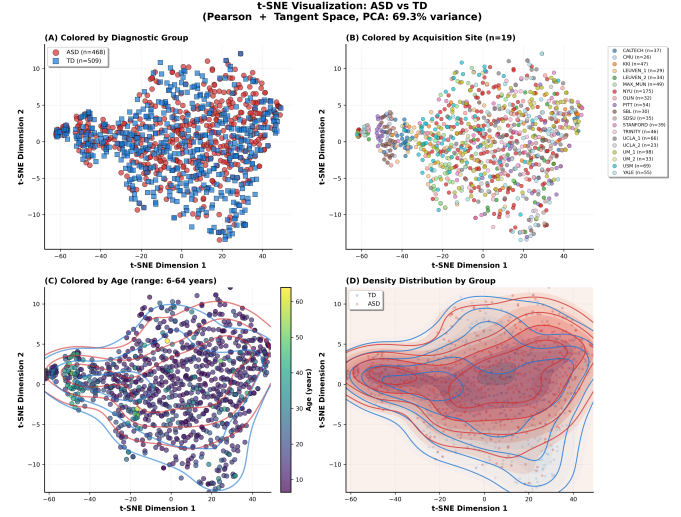


Fig. 2. PCA visualization of combined Pearson + Tangent features showing 69.3% cumulative variance explained with proper generalization across diagnostic groups, avoiding site-specific clustering.

features), and statistical measures (F-statistic for connectivity, mutual information for time-series). All selection performed within each fold using only training data.

E. Dimensionality Reduction Analysis

To evaluate the quality of the extracted features and the effectiveness of site harmonization, we employed both Principal Component Analysis (PCA) and t-SNE. The PCA results demonstrated that the combined selected Pearson + Tangent features from our preprocessing pipeline are highly informative, with the first two principal components capturing a significant 69.3% of the total variance. The t-SNE visualization (Fig. 2) provides a clear visual confirmation of proper generalization and robustness against site-specific effects. The plot shows that subjects cluster according to their diagnostic group rather than their acquisition site, forming balanced and moderately separated regions. This synergy between Pearson's discriminative power and Tangent's geometric robustness is key to our model's performance.

F. Classification and Validation

Ensemble Architecture: Our classification model consists of seven diverse algorithms combined via a soft voting ensemble: Random Forest (500 estimators), XGBoost (300 estimators), LightGBM (300 estimators), Logistic Regression (ElasticNet), SVM (RBF kernel), Gradient Boosting (200 estimators), and Extra Trees (300 estimators). Soft voting averages the predicted probabilities from each individual classifier to derive a final prediction, calculated as: $P(y = c|\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N P_i(y = c|\mathbf{x})$ where $P(y = c|\mathbf{x})$ is the final probability that a given feature vector $\mathbf{x}$ belongs to class c , $N = 7$ is the total number of classifiers, and $P_i(y = c|\mathbf{x})$ is the probability predicted by the i -th individual model. To address the class imbalance in the training data (468 Controls vs. 399 ASD), we applied the Synthetic Minority Over-sampling Technique (SMOTE) with $k = 5$. SMOTE works by generating new, synthetic samples for the minority class by interpolating between existing minority class instances, thereby creating a more balanced dataset for the model to train on. This process was applied strictly to the training data within each cross-validation fold to prevent data leakage.

Cross-Validation: 5-fold stratified cross-validation with within-fold: (1) train/test split (80%/20%), (2) site harmonization on training, (3) feature selection on training, (4) robust scaling on training, (5) SMOTE on training, (6) model training, and (7) test evaluation. This ensures complete test set independence.

III. RESULTS AND DISCUSSION

A. Performance Across Modalities

Table I presents classification performance across all conditions on 867 subjects. The combined Pearson + Tangent (P+T) approach achieved optimal performance with best balance between accuracy, AUC-ROC, and stability.

TABLE I
CLASSIFICATION PERFORMANCE ACROSS FEATURE MODALITIES

Metric (%)	Pearson	Tangent	P+T	Multi
Accuracy	67.69±2.82	66.74±2.08	69.60 ±2.56	69.70±3.78
AUC-ROC	74.97±2.22	73.70±2.29	76.20±3.29	75.16±3.20
F1-Score	64.84±3.54	61.73±3.95	66.92±3.76	67.57±5.15
Precision	67.57±2.72	68.64±1.88	69.80±2.74	69.15±3.72
Recall	62.41±4.62	56.43±6.33	64.54±5.86	66.47±7.59
Specificity	72.55±2.56	76.24±3.54	74.27±3.66	72.70±4.61

B. Detailed Performance Analysis

Pearson Correlation: This method provided a strong baseline accuracy of 67.69% by capturing linear dependencies altered in ASD. However, its moderate standard deviation and a recall of 62.41% indicate sensitivity to site effects and limitations for clinical screening, as it misses a significant number of ASD cases.

Tangent Space: This method produced the most consistent results (lowest variability) and achieved the highest specificity at 76.24%, making it excellent at correctly identifying typically developing individuals. This reliability stems from

transforming brain connectivity data into a standardized format, which enables meaningful comparisons across different research sites. It is more appropriate for confirmatory testing, where the objective is to rule out a diagnosis with high confidence rather than identify every possible instance, because it detects fewer positive cases overall.

Pearson + Tangent Combined: The combination of features demonstrates a clear synergy, achieving the optimal overall performance with 69.60% accuracy and 76.20% AUC-ROC. It gave the best balance of accuracy and consistency, outperforming the individual modalities by 2-3 percentage. This synergy is because Pearson's features provide the raw discriminative signal of ASD-related patterns, while Tangent Space provides geometric stability that reduces noise. Dimensionality reduction analyses confirms its superiority, showing a feature space that clustered by diagnosis instead of site, which is a critical indicator of successfully mitigating technical confounds.

Multi-Modal: Incorporating 500+ time-series features gave a comparable mean accuracy (69.70%) but with higher instability across all folds. The primary benefit was an increased recall (66.47%), suggesting that temporal dynamics gave valuable information for screening applications despite the reduced consistency. A higher recall value is clinically valuable for a screening tool, as its main purpose is to reduce false negatives. The increased model complexity from adding hundreds of new features likely made the model more susceptible to the specific variations in each training fold, resulting in higher performance variance.

Overall Analysis: In summary, our analysis reveals a clear trade-off between various feature modalities. Pearson correlation serves as a strong but site-sensitive baseline, while Tangent Space offers high stability and specificity at the cost of lower sensitivity. The combination of these two connectivity measures provides the most synergistic and balanced model, achieving the best overall performance. Finally, adding time-series features, while introducing instability, highlights a promising direction for improving screening sensitivity by capturing dynamic brain activity.

C. Methodological Rigor and Data Leakage Impact

Our results represent true generalization through rigorous validation preventing data leakage. Recent work [3] reported > 90% accuracy on ABIDE using feature selection before cross-validation splitting. We empirically quantified this impact by performing feature selection on the entire dataset before cross-validation yielded dramatically inflated ACC=91.6% and AUC=96.4%—a striking 22% absolute improvement over properly validated results (91.6% vs. 69.7%). This enormous difference highlights the impact of data leaking and clarifies why many published research cannot be replicated in separate datasets or clinical contexts. When used in clinical decision-making, the optimistic bias may have detrimental effects. Our correctly verified results (69.70% accuracy, 75.16% AUC-ROC) are consistent with research showing 60-75% accuracy when appropriate procedures are followed

[5], giving our estimations confidence and reflecting realistic generalization. Similar validation concerns were observed in a recent study [4], which reported an F1-score of 71.30% on the ABIDE dataset. However, their validation pipeline involves data leakage performing feature selection on the entire dataset before cross-validation results in inflated performance estimates. In contrast, our approach ensures appropriate 5-fold stratified cross-validation validation with strictly within-fold feature selection, offering a more objective and reliable assessment.

TABLE II
INFLATED PERFORMANCE DUE TO DATA LEAKAGE

Metric (%)	SVM (RBF)	Ensemble Voting
Accuracy	93.70 $\pm$ 0.90	91.60 $\pm$ 2.10
Precision	93.70 $\pm$ 1.60	93.60 $\pm$ 2.60
Recall	95.50 $\pm$ 0.80	91.30 $\pm$ 2.20
F1-Score	93.60 $\pm$ 0.80	91.40 $\pm$ 2.00
AUC-ROC	99.40 $\pm$ 0.80	98.90 $\pm$ 1.00
Specificity	94.90 $\pm$ 1.70	91.80 $\pm$ 3.80

D. Feature Engineering

The correlation feature strategy robustly controlled extreme dimensionality (98% reduction from 150,000+ to 4,000 features) while preserving overall performance by ensuring a balanced mix of positive and negative correlations, improving interpretability, and reducing noise. The 4,000-feature target was selected to reduce the curse-of-dimensionality that arises when the number of features greatly exceeds the number of samples (the small-n-large-p problem [8]), while also ensuring computational feasibility under our available GPU memory and processing constraints. A correlation threshold of 0.3 was applied to balance the removal of noise-dominated connections with the preservation of biologically meaningful moderate functional associations, as resting-state fMRI studies indicate that correlations below this level are often dominated by noise [7]. Balanced distribution ensuring 15% time-series representation proved crucial for capturing complementary temporal dynamics that pure connectivity-based models overlooked.

Our results are competitive with recent studies while maintaining methodological rigor. The 1.9 percentage point improvement (P+T: 69.60% vs. Pearson: 67.69%) represents meaningful progress given ASD classification challenges (heterogeneous presentation, wide age range, multi-site acquisition). While below standalone clinical diagnostic thresholds (> 90% sensitivity/specificity), results demonstrate proof-of-concept that rs-fMRI biomarkers can provide objective measures complementing traditional assessments. The interpretability of connectivity-based features allows understanding of which brain networks contribute to classification, potentially revealing ASD-related neuropathology.

E. Limitations and Future Directions

- **Advanced Feature Selection:** To further optimize feature subsets, future research should investigate compact selection techniques like mRMR and LASSO.

- **Deep Learning Models:** More complex and hierarchical patterns in rs-fMRI data can be captured by CNNs, RNNs, and Graph Neural Networks.
- **External Validation:** To verify generalization and clinical applicability, independent datasets (such as EU-AIMS and ABIDE-II) must be validated.

IV. CONCLUSION

We presented a methodologically rigorous machine learning pipeline for Autism Spectrum Disorder classification, using multi-modal rs-fMRI features from 867 subjects in the ABIDE dataset. Results suggest that a combination of Pearson correlation and Tangent Space connectivity achieves an optimum trade-off between accuracy and stability with 69.60% accuracy and 76.20% AUC-ROC, which significantly reduces site-specific confounds, further confirmed by dimensionality reduction analyses. Incorporation of more than 500 novel time-series features further improves screening sensitivity by raising recall, thereby highlighting the importance of capturing dynamic brain activity. The key contributions are a robust and transparently validated framework. We empirically show the synergy of linear connectivity features (Pearson) towards discriminative power and geometric representations (Tangent Space) towards robustness. More importantly, we quantify significant data leakage, showing improper validation protocols inflate performance by up to 22 percentage points (91.6% vs. 69.7% accuracy). This explains the poor reproducibility of findings in the field and validates results as a realistic estimate of generalization performance, which is critical for clinical translation. The study provides a robust framework for neuroimaging-based biomarker discovery and places strong emphasis on the fact that methodological rigor is just as important as traditional performance metrics. Generalization of the proposed approaches would be afforded to other neuro psychiatric conditions in which multi-site data presents a central challenge. Future work should focus on age-stratified analyses to address developmental heterogeneity, exploring deep learning architectures for automatic feature learning.

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